

The AI Dependency Index (IDA): Quantifying Minimum Workforce Continuity in AI-Dependent Organizations

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Abstract

Problem. Organizations adopting artificial intelligence compute how many people they can release. We did not find a published framework that quantifies how many they must retain to keep operating if the provider becomes unavailable. Two developments in 2026 make that second number urgent: the price of inference is not anchored to any verifiable cost structure, and on 12 June 2026 a United States export-control order required a frontier provider to suspend two of its most advanced models for all foreign nationals; unable to verify nationality per user, the provider disabled both models for every customer worldwide until the controls were lifted on 30 June [5]. The disruption originated in policy, not infrastructure. **Gap.** Governance and operational-resilience frameworks prescribe dependency mapping, criticality tiering and fallback protocols. The AI Resilience Framework [1] extends impact-tolerance constructs to AI dependencies and identifies a quantitative substitutability score as an unrealised extension, which this work develops in workforce terms. **Method.** The AI Dependency Index (IDA) is a deterministic model built from six operational inputs managers already hold. It returns four outputs in one accounting: a retained workforce, a risk-adjusted saving, a maximum defensible investment, and a bounded 0–100 exposure score discounted by verified external backup. **Findings.** Past the continuity floor, further automation raises the AI bill without releasing anyone: savings fall while exposure rises. Under fixed-cost architectures savings plateau instead of falling, yet exposure still rises, so automation then buys dependency for nothing. Both behaviours follow from the equations. **Availability.** The model, its formulas, an executable implementation and a battery of 24 edge cases are published under the MIT licence and deposited with a permanent identifier [2]. It runs at modelo-ida.web.app.

Keywords: operational resilience; business continuity; AI dependency; vendor concentration; workforce planning; decision support systems.

1. Introduction

Every business case for automation answers the same question: how many people will this let us do without. The question is old, the arithmetic is familiar, and the answer feeds directly into a return-on-investment calculation that boards know how to read.

There is a second question that the same business case does not ask. If the automated work now depends on an external provider, and that provider stops being available, how many people does the organization need in order to keep operating at all? This is not the first number with the sign reversed. It is a different quantity, governed by a different constraint, and in current practice it is simply not computed.

The asymmetry was tolerable while the risk was hypothetical. Two developments in 2026 make it concrete.

The first is price. The amount an organization pays per unit of inference is not anchored to a verifiable cost structure that the buyer can inspect or predict. It is set unilaterally by providers whose own business models are not yet settled, which means the operating cost of an automated process can move for reasons entirely outside the adopter's control and unrelated to its own consumption.

The second is availability, and it is no longer hypothetical. On 12 June 2026 the United States Department of Commerce issued an export-control directive to Anthropic requiring it to suspend access to two of its most advanced models, Claude Fable 5 and Claude Mythos 5, by any foreign national, whether inside or outside the United States [5]. Unable to verify the nationality of each user in real time, the provider disabled both models for every customer worldwide [6]. The controls were lifted on 30 June, nearly three weeks later, and access was restored in stages [7]. No infrastructure failed. No service-level agreement was breached in the ordinary sense. No multi-cloud architecture would have helped, because the constraint was not technical.

An organization that had automated a critical process on that provider had, for those three weeks, exactly as much operating capacity as it had retained people who could still do the work by hand. Very few of them knew that number in advance.

Provenance. The model was not derived in the abstract. It condenses roughly a decade of consulting practice on AI and automation adoption across regulated and operationally critical sectors, during which its formulation was repeatedly revised against engagements that cannot be identified here for reasons of client confidentiality. That history is the reason the model asks for six inputs rather than sixty: every input that a manager could not answer from memory in a meeting was removed. It is also, necessarily, a limitation on the evidence this paper can offer: the practice informed the design but is not reported here as data, and the published case in Section 4 is constructed rather than observed. Independent empirical validation therefore remains open, and Section 6 states what would close it.

Contribution. This paper proposes the AI Dependency Index (IDA), a deterministic model that computes the minimum workforce an organization must retain for continuity, the saving that remains once that constraint and architectural fragility are priced in, the maximum investment such a project can justify, and a bounded index of operational exposure. The model is deliberately small: it consumes six inputs that any operations manager already possesses, requires no dataset and no training, and is published in full so that its results can be reproduced or refuted.

2. Related work and the gap

Three bodies of work bear on the problem, and none of them returns a headcount.

Governance standards. ISO/IEC 42001 establishes requirements for an AI management system, including risk assessment and treatment. The European Digital Operational Resilience Act (DORA) obliges financial entities to map information and communication technology dependencies and to test resilience against severe but plausible scenarios. Both are prescriptive about process and silent about magnitude: they require that a contingency plan exist, not that it be sized.

Business continuity practice. Established continuity methodology already provides the conceptual apparatus for a minimum operating level, most directly through business impact analysis and the notion of a minimum business continuity objective. What it does not provide is a way to derive that level from the specific structure of an AI dependency, where the substituting resource is human labour that the automation decision itself is in the process of removing.

Recent AI-resilience literature. The closest published work is the AI Resilience Framework [1], which argues that governance centred on trustworthiness leaves an obligation of resilience inadequately covered, and proposes dependency mapping, tiering by criticality and substitutability, fallback protocols, and management of provider concentration. That framework explicitly identifies a quantitative substitutability score as a natural extension that would allow tiering thresholds to be calibrated rather than asserted, and does not develop it. The present work takes up that extension and expresses substitutability in the only unit in which a fallback protocol can actually be executed: people who can do the work.

Adjacent index work exists but answers a different question. Firm- and industry-level indices of AI transformation measure opportunity, disruption risk and value creation; they describe how much an organization stands to gain, not how much operating capacity it would retain if the technology became unavailable.

Across those three bodies of work we did not find a published framework that returns a headcount. The prescription is qualitative, while the decision a manager has to sign is quantitative.

3. The model

3.1 Inputs

The model takes six operational quantities. The design constraint was that each must be answerable by an operations manager without a data project.

Symbol	Input	Unit
P	Headcount of the area under analysis	people
N	Monthly payroll of that area	currency/month
h_p	Working hours per person per month	hours
a	Share of the work that AI could perform	fraction
r	Share of that work requiring mandatory human sign-off	fraction
η	Effectiveness actually achieved by the AI	fraction

Table 1. Operational inputs. Three further parameters describe the project rather than the operation: initial investment I, monthly maintenance M, and the approval horizon $T_{\max}$. A tenth input, external backup R, is introduced in Section 3.6.

3.2 Effective automation

The share of work that is genuinely automatable is smaller than the share that is technically automatable. Two independent discounts apply: the fraction of that work which regulation or policy requires a human to authorise, and the effectiveness the system actually achieves in production rather than in demonstration.

$$a_{ef} = a \cdot (1 - r) \cdot \eta$$

The composition is multiplicative, and the resulting gap is not a rounding error. An organization that believes it is automating 60 per cent of a process, of which one fifth requires sign-off, with a system performing at 80 per cent, is in fact automating 38.4 per cent. Every subsequent quantity is computed on a a_{ef} , never on a.

3.3 The continuity floor

The organization declares $H_{\min}$, the hours per month that must be covered whatever happens: the degraded mode it is not permitted to fall below. This is a policy or regulatory statement, not an estimate derived from the model. The floor in people follows directly.

$$P_{\min} = \text{ceil}(H_{\min} / h_p)$$

3.4 Retained workforce

Economics alone would set headcount by the share of work still done by people. Continuity sets a lower bound that economics may not cross. The retained workforce is the larger of the two.

$$P_{econ} = \text{ceil}(P \cdot (1 - a_{ef})) \quad P' = \max(P_{econ}, P_{\min})$$

The maximum operator is where the model departs from a conventional business case, and it is the source of every non-obvious result in Section 4. When $P_{\min}$ binds, additional automation stops converting into released people while continuing to convert into cost.

A boundary case deserves explicit treatment. If $P_{\min} > P$, the organization is already below the headcount required to operate without the AI. The model reports this rather than clamping it: it is a finding about the present state, not an error in the inputs.

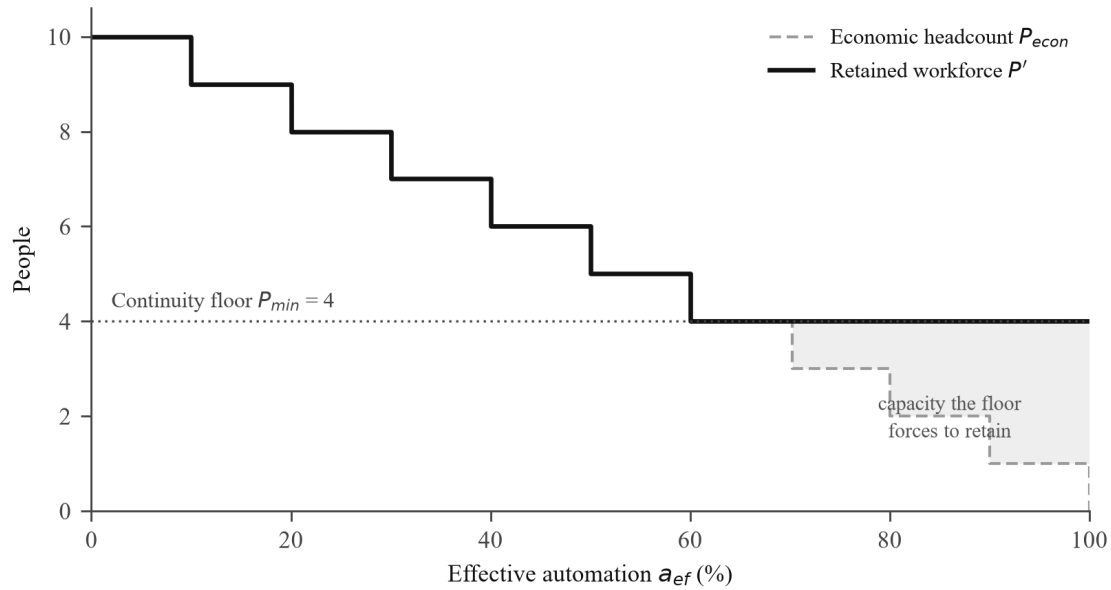


Figure 1. The continuity floor. Economic reasoning alone would keep reducing headcount as effective automation rises (dashed). The retained workforce (solid) follows it only until it reaches the floor the organization declared, and then stops. The shaded area is the labour capacity that continuity forces the organization to keep, and that a conventional business case would have released. Reference configuration of Section 4.2.

3.5 Risk multiplier and risk-adjusted saving

Architectural fragility is priced as a surcharge on the AI expense. The multiplier starts at unity and accumulates one premium per declared fragility.

$$\lambda = I + \sum \rho_i$$

Condition	Premium ρ
AI runs on the organization's own infrastructure	0.02
AI runs across multiple cloud providers	0.10
AI runs on a single cloud provider	0.30
No documented contingency plan	+0.20
Sensitive data processed outside the organization	+0.15

Table 2. Architecture premiums. The first three are mutually exclusive; the last two are additive. The attainable maximum is $\lambda = 1.65$. These coefficients are declared expert judgement; see Section 6.

Writing C for the monthly AI expense, the risk-adjusted saving subtracts the surcharged AI cost and maintenance from the payroll of the people actually released.

$$S = (P - P') \cdot (N / P) - C \cdot \lambda - M$$

S is an economic evaluation quantity, not a cash-flow projection. The organization does not write a cheque for $C \cdot \lambda$; the surcharge exists so that a fragile architecture appears expensive while it is still cheap to change. An accountant will compute a larger saving, namely the same expression with $\lambda = 1$, and both numbers are correct for their respective purposes.

3.6 Payback, investment ceiling and external backup

Simple payback follows as $T = I / S$. Inverting it against the horizon at which the organization approves investments yields the quantity that has proved most useful in practice.

$$I_{max} = S \cdot T_{max}$$

This reframes the decision. Rather than reporting that a project pays back in twenty months and inviting a binary approval, the model states the maximum amount the project can justify, which is a negotiating position computed from the organization's own saving and its own risk.

Exposure is discounted by verified external backup. R is the number of people outside the analysed area who can perform the work on the day of an outage, without training.

$$coverage = \min(R / P_{min}, 1) \quad (coverage = 0 \text{ when } P_{min} = 0)$$

Two design decisions constrain R. First, it is admitted only when three conditions are jointly declared: that those people have performed this process, that the manual procedure is documented, and that the handover has been rehearsed. If any is absent, R is forced to zero. R is the only input that lowers the reported risk, and is therefore the one most exposed to optimistic reporting. Second, coverage discounts exposure but never eliminates it: the maximum discount is 0.7, because people brought in from elsewhere abandon their own posts, do not perform the task daily, and lack accumulated context.

It should be noted that R does not reduce P'. External backup lowers how exposed the organization is, not how many people it needs.

3.7 The index

The index composes the three factors that determine exposure: how much of the work now depends on the system, how fragile the architecture carrying it is, and how much of the continuity floor is externally covered.

$$IDA = \min(a_{ef} \cdot \lambda \cdot (1 - coverage \cdot 0.7) \cdot 100, 100)$$

The bounds are interpretable: 0 denotes no measurable dependency, 100 denotes the maximum dependency the model represents. The cap is required because the unbounded product reaches 165 at $a_{ef} = 1$ with $\lambda = 1.65$; everything above 100 is already maximal exposure, and truncating preserves a stable meaning for the scale. Reported bands are 0–30 resilient, 30–60 watch, 60–100 critical.

Saving and exposure are reported separately and are not combined into a single score. An organization may hold a large saving with high dependency, or a small saving with low dependency; collapsing the two would conceal precisely the trade-off the model exists to expose.

4. Properties and worked example

4.1 Reference case

A pharmacy chain automates customer enquiries. The area has ten people, a monthly payroll of 16,000, and 160 working hours per person. Sixty per cent of the work is automatable, of which twenty per cent requires pharmacist sign-off; the system performs at eighty per cent. Six hundred and forty hours per month must be covered whatever happens. The project costs 50,000 with 500 per month of maintenance, runs on a single cloud provider with no contingency plan, and is evaluated against an eighteen-month horizon.

Quantity	Value
Effective automation a_{ef}	0.384
Continuity floor P_{min}	4 people
Economic headcount P_{econ}	7 people
Retained workforce P'	7 people (3 released)
Risk multiplier λ	1.50
Risk-adjusted saving S	2,456.80 per month

Payback T	20.4 months
Investment ceiling $I_{\max}$	44,222
Index IDA	58 (watch)

Table 3. Reference case. The decisive line is the last but one: the project was presented at 50,000 and the model states that it is worth at most 44,222 to this organization at this level of risk.

4.2 An automation optimum exists under variable cost

Setting $r = 0$ and $\eta = 1$ to isolate the effect, and letting the automated share rise, produces a result that contradicts the intuition the business case is built on.

Effective automation	Naive promise	Actual saving	Payback	IDA
60% (optimum here)	6,220	6,220	8 months	90
80%	8,460	5,260	10 months	100
100%	10,700	4,300	12 months	100

Table 4. Past the continuity floor the organization becomes poorer and more dependent simultaneously. The naive column assumes every automated hour is a salary removed.

The mechanism is the maximum operator of Section 3.4. Once P_{econ} falls below P_{min} , the floor binds: released headcount stops growing while automated hours, and therefore the AI bill, continue to grow. Savings decline. The location of the optimum is case-specific, moving with headcount, wages, unit cost and the declared floor; the shape is general.

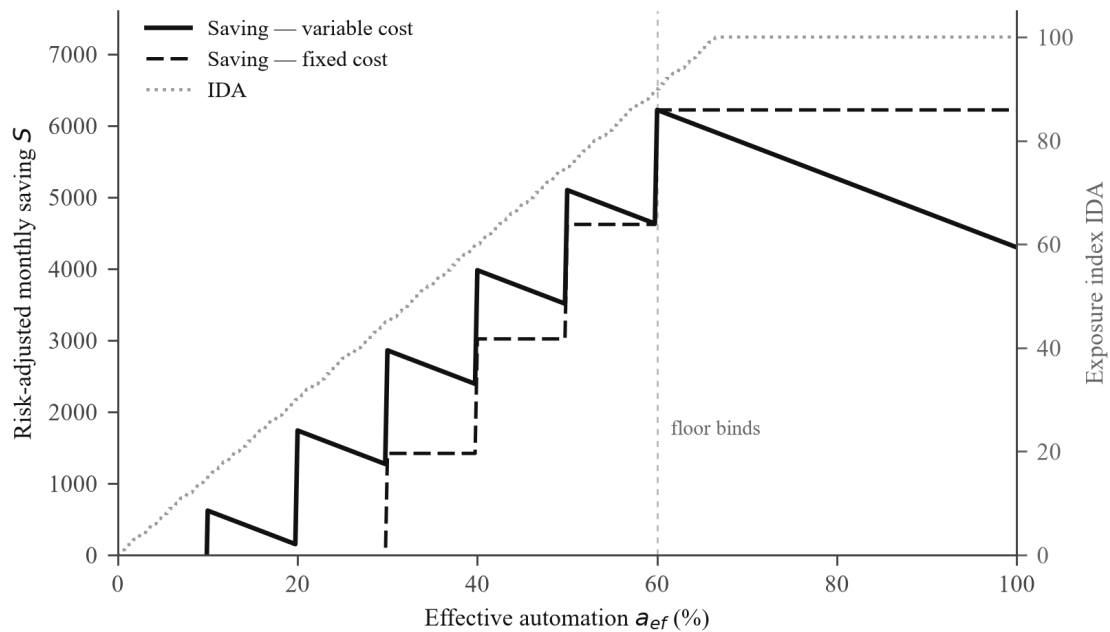


Figure 2. Saving against exposure. Once the floor binds at 60 per cent, the two cost structures diverge: under variable cost the saving declines, because automated hours keep accruing while released headcount does not; under fixed cost it plateaus. The exposure index rises in both cases. To the right of the dashed line, further automation buys no additional saving in either regime and continues to buy dependency. The fixed cost is set to the variable-branch expense at $a_{\text{ef}} = 0.60$, so that both curves coincide there and the comparison isolates the cost structure rather than the level of spend.

4.3 Fixed-cost architectures plateau rather than decline

The preceding result assumes the AI is billed per unit of use. Where the organization operates its own infrastructure, C is a constant, and since released headcount is non-decreasing in a_{ef} while every other term is fixed, S is non-decreasing. In the same configuration as Table 4 with a fixed monthly cost, the saving is

identical at 60 and at 100 per cent effective automation.

The consequence is not that the trade-off disappears but that it changes currency. Under fixed cost, automating past the floor no longer makes the organization poorer, yet the index still rises from 90 to 100 across the same interval. Additional automation purchases dependency for nothing. Under variable cost the organization loses money and resilience together; under fixed cost it loses only resilience, which is harder to see in a set of accounts and more expensive to repair afterwards.

The distinction this makes visible is worth stating plainly: **automation does not imply autonomy**. An organization can automate an increasing share of its work and, at the same time, become less able to operate on its own. The two quantities move independently, and only one of them is usually measured.

4.4 Uncertainty

Point estimates of a , r , η and unit cost are rarely defensible at the moment a project is approved. A Monte Carlo layer draws from ranges the analyst declares and reports the probability that the investment is recovered within $T_{\max}$, together with the probability of a negative saving and of late recovery. In the reference case, sampling plausible ranges returns a recovery probability materially below one half, which is itself the finding: at these ranges the project is closer to a wager than to a plan, and the lever that moves the number is rarely more automation but lower architectural fragility or a smaller initial investment.

5. Verification and reproducibility

The model is published together with an executable implementation that runs entirely in the browser, with no server component and no transmission of the data entered. Any reader can therefore reproduce every figure in this paper without installing software, registering, or exposing operational data to a third party. The implementation is available at <https://modelo-ida.web.app> and its source at github.com/All-Aideas/Modelo-IDA.

Correctness is checked by a battery of 24 edge cases distributed with the source. The battery does not compare the engine against itself: its expected values are computed by an independent re-implementation of the formulas exactly as published, so that any divergence between the documented model and the running code fails the test. The cases cover the boundaries where models of this kind break, including a floor exceeding current headcount, a single-person area, a thousand-person area, division by zero in the coverage term, an AI hour costing more than a human hour, both extremes of λ and of the approval horizon, declared but unverified backup, and both cost structures of Section 4.3.

The implementation, the formulas, the test battery and this document are released under the MIT licence and deposited with a permanent identifier [2].

6. Limitations

Four limitations bear on how the results should be read, and are stated here rather than left for the reader to discover.

The coefficients are expert judgement. The premiums of Table 2 and the 0.7 discount cap were set from practice, not calibrated against a dataset of incidents. They are design parameters of the model. The immediate consequence is that comparisons within a single architecture are meaningful while absolute values across very different organizations should be treated with caution. Calibration against a corpus of documented provider outages is the most valuable extension available and the one the authors regard as the principal item of future work.

Released hours are converted into eliminable positions. The economic branch assumes that hours the AI removes translate into people the organization stops needing. In practice an organization may reassign those people, absorb growth without hiring, or reduce overtime instead of headcount. In all of those cases the saving exists but does not take the form the model gives it. P' should therefore be read as labour capacity no longer

required, not as a redundancy instruction, and where reduction is not intended S is the ceiling of what could be saved rather than what will be.

The multiplier is applied to AI spend, not to outage impact. λ surcharges the cost of the AI because that is a figure the organization holds, whereas the cost of non-operation generally is not. This produces a counter-intuitive artefact across architectures of very different cost: a safer, expensive on-premises deployment can carry a larger absolute premium than a fragile, cheap API. Within one architecture the comparison is sound, which is how the model is used; across architectures the index rather than the monetary premium is the quantity to compare. Anchoring the premium to an hourly cost of non-operation is the most important pending refinement.

Scope. The model addresses cost and operating continuity. It does not model revenue lost during an outage, reputational damage, regulatory penalties, or the risk that a provider degrades a model without withdrawing it, which the June 2026 episode suggests deserves separate treatment. Two organizations with the same IDA can therefore face very different losses from the same interruption: the index measures exposure to an outage, not the cost of one.

7. Conclusion

The organizations that were unable to operate for three weeks in June 2026 did not suffer an infrastructure failure. They discovered, at the worst possible moment, a number they had never computed. This paper proposes that the number be computed in advance, using inputs the organization already has, and reported alongside the saving rather than instead of it.

The model reframes the human factor from a declared organizational value into measurable continuity infrastructure, and gives decision-makers a single defensible figure to bring to a vendor negotiation. Its principal claim is modest and testable: that past a floor which the organization itself declares, additional automation stops buying savings and continues buying exposure, and that both behaviours follow from the equations rather than from narrative.

The model is offered as a proposed quantitative framework, not as a validated index. Its coefficients await calibration and its predictions await empirical cases. Both the implementation and the test battery are public precisely so that the refutation, if it comes, can be specific.

The architecture premiums of Table 2 and the backup discount are the first quantities that should change. Practitioners who run the model on their own operation, and organizations able to contribute anonymised incident data, are invited to write to the author: calibrating those values against observed cases is what would move the IDA from a proposed framework to a validated one.

Availability

Interactive model: <https://modelo-ida.web.app> · **English edition:** <https://modelo-ida.web.app/en>

Source code and test battery: <https://github.com/All-Aideas/Modelo-IDA>

Archived, all versions: <https://doi.org/10.5281/zenodo.22070415> · **this version:** <https://doi.org/10.5281/zenodo.22070416>

Licence: MIT. No registration, no server, no data leaves the browser.

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